

Data-Driven Analysis of EV Energy Prediction and Planning of EV Charging Infrastructure

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Abstract—To reduce the current crisis on energy and environmental problems, the adoption of electric vehicles is an effective way. But limited access and availability of charging infrastructure are restricting the EV market development. This paper presents data-driven analysis of current EV charging infrastructure to plan future EV infrastructure in the city. This paper tackles the problem using three approaches. The first approach that is covered in this paper is to predict the number of EV charging stations that need to be implemented in a geographic location. This is done using the SARIMAX model on the California EV charging station dataset that is preprocessed to be split into zip codes. The second approach that is tackled by this paper is the energy consumption prediction by an EV charging station. This information is useful to allow planners to understand how much energy is consumed by an EV station across multiple different time periods. An optimized Meta-Fusion regression model based on a stacked model is proposed for this approach and is applied to the Palo Alto charging station dataset. The third approach utilized the K-means clustering algorithm on the Palo Alto charging station dataset to predict the optimal location for EV charging stations in that city. The SARIMAX mode had an R2 score of 0.97, a MAE of 6.50, and RMSE of 17.43 when predicting EV station counts in the zip code 92101. The stacked model had an R2 score of 0.98, a MAE of 0.91 and an RMSE of 2.26 when predicting the daily energy consumption by an EV station.

Keywords—Big Data, EV Charging Infrastructure, EV Charging, Data-Driven Analysis, EV Electricity Prediction, Machine Learning.

I. INTRODUCTION

Climate change is one of the most pressing issues in the world today. This issue is mainly due to the large amounts of carbon dioxide and other greenhouse gases (GHG) being let into the air. According to the NIEHS [1], air pollution causes over 6.5 million deaths each year globally. According to the WHO, this is because 99% of the population in the world is breathing in air that exceeds their limits on breathable air. One of the major contributors of GHG is emissions from road transportation. Road transportation uses a lot of fossil fuels and emits GHG when the fuels are converted into energy. According to Gurwa et. al (2019)[2], 27% of the world's GHG

emissions come from road vehicles. The biggest way to solve this specific problem is for consumers to start buying electric vehicles (EVs) because EVs are battery powered and do not emit any GHGs when generating energy to move.

However, there are a few problems that act as barriers for the growth of electric vehicles. One of the main problems that must be addressed as people start switching to EVs is to implement better EV infrastructure in the cities and near highways, because people need to charge their vehicle in the middle of a long trip just as they would be refueling a gasoline powered vehicle. A positive to EVs is that most of them are sold with a Level 1 charger that can be plugged into an outlet at home, so some people can use this to charge at home. But this is the slowest method of charging and is not useful for long road trips. Another problem with charging EVs is that it takes much more time to charge an electric vehicle to full capacity than it takes a gasoline-powered vehicle to refuel at a gas station. There are currently three types of EV chargers, and the following table shows the features of each type. The variations in the times listed in the table below are caused by the different battery types used in each vehicle. In our project, we are primarily focusing on Level 2 and DC fast charging stations because those are the fastest type of chargers available.

TABLE I
OVERVIEW OF ELECTRIC CHARGES (USDOT)

Type	Level 1	Level 2	Level 3(DC Fast)
Power	1 kW	7 - 19 kW	50 - 350 kW
Time Taken to Charge a Battery from 0% Capacity	40 - 50 hours	4 -10 hrs	20 - 60 min (0 to 80% cap)
Range per Hour of Charge	2 - 5 miles	10 - 20 miles	180 - 240 miles
Typical Locations Placed	Home	Home , Office , Public	Public

The above problems motivated us to come up with a solution for the planning of EV infrastructure with an AI powered model using Machine Learning and Deep Learning Techniques. The main objective of this research is to create a model that can optimally predict where EV infrastructure can be placed in a city. The model will take into consideration various factors related to EV charging such as the location of current stations, EV energy consumption.

This research also considered the weather data to understand how EV charging demand is affected by weather. The optimal EV infrastructure network can cater to the charging needs and reduce the life cycle costs of the transportation system. Recent advancement in machine learning helps analyze present data and come up with efficient solutions that solve the planning of charging infrastructure in a city. It can help in planning and management such as prediction of location of new charging infrastructure, demand for charging and scheduling. Hence, knowledge of machine learning models and AI is a key solution for enhancing the sustainability of EV transportation. Our project can be used for a few applications. We expect other cities around the world to expand on our model and use it to build new EV infrastructure in their cities. Another application that can be created with the help of our model is a mobile application that can be used to help owners to get the various charging options when they don't have access to charging stations.

This paper is divided into the following sections. Section II describes a literature survey of previous related work. Section III provides an overview of the technical survey and prior art. Section IV gives an overview of the Methodology, which is subdivided into Dataset, Data Preprocessing which is again subdivided as Data Cleaning and Data Merging. Section V discusses the machine learning model used in the project for load forecasting, forecasting of energy consumption and location recommendation. Section VI details the results obtained by the machine learning model. Section VII describes future improvements to the project and concludes with a summary of the project.

II. RELATED WORK

This section provides an overview of the research conducted for this project, encompassing an extensive review of the existing literature and technology survey.

A. Literature Survey

In the literature survey conducted, there are many different approaches that authors use to predict new EV charging infrastructure in a city. One of the methods used is to figure out the charging demand of EV stations in order to understand the energy consumption that a new EV charging station will require. [13] used Pattern Sequence forecasting, [19] used an GRU model and [16] used an LSTM model with some preprocessing and tuning methods to solve this problem. [6] analyzed the characteristics of EV charging demand and identified the factors that affect it, such as weather conditions, time of day, and location. [20] and [24] have each compared

three different ensemble learning models in order to figure out which model predicts the energy consumption of a Level 2 EV charging station. The former found that the XGBoost model and the latter found that a Coarse Gradient-SVM model performs the best out of the models that they tested. Some papers looked at the user's charging behavior, such as figuring out when the station is empty or predicting when users will use the station. [10] uses a GCN and a GRU whereas [25] uses a regression model on streamed data to predict station availability.

Another approach used is to directly predict the locations of the new EV charging stations in an area. [7] used features from EV charging data in order to create a location-based energy demand risk-level. [9] used a location allocation model to predict new EV stations' location. [8] and [11] incorporated location based features of the EV station location dataset with the location-allocation model and XGBoost model, respectively. There are a few authors that used uncommon methods to predict these locations. [12] used location information of the EV stations and its surrounding area to rank the popularity of the stations. [14] used the Markov chain principle and a particle swarm algorithm on people's travel data and [17] used a graph convolutional network on traffic data. [22] uses two optimization algorithms and then a decision making algorithm and [23] uses a game theoretical framework to solve this problem. [21] uses a decision support system to predict EV station locations for electric taxis. [27] uses cost constraints provided by private enterprises in their location-based EV prediction.

Table II here gives a list of papers we researched, and it includes their objectives, Region, different types of datasets they used, models, and evaluation metrics.

B. Technology Survey

There are few companies that have created a software to plan EV charging infrastructure around various places in the United States. Almost all these software are not available to the public and are proprietary. We are not able to get information on how they are implemented. The National Renewable Energy Laboratory (NREL) created one of these tools called Electric Vehicle Infrastructure – Projection Tool (EVI-Pro)[28].

The website states that users can use the software to analyze travel patterns, charging demand, and design the infrastructure capable of meeting demand. This tool was used to help analyze the requirements of EV infrastructure planning in the entire state of California. The US Department of Energy has a tool called EVI-Pro Lite which is based on EVI-Pro and is available for public use[29]. This is a web tool that allows users to plan the number of charging stations required in a city or state. This tool has few pre-selected urban areas. Once you select the location, then you enter in the number of vehicles that you want to support there, and then click on the Calculate button. It will do some calculations and show you the number of Level 2 and Level 3 charging stations that need to be implemented

TABLE II
COMPARISON OF EXISTING RESEARCH PAPERS

Ref	Objective	Region	Dataset	ML Models	Deep Learning	Other Models	Eval Metrics
[6]	Calculating the daily charging demand of EV stations based on weather and traffic data	South Korea	Traffic Weather	N	N	Forecasting model with big data technologies P-median	Compared to standard models
[7]	Calculate the energy demand risk level of an EV charging station wrt to Charging event and station data.	United Kingdom	Charging Event Charging Station	Clustering	N	N	Corelation Coefficient
[8]	Include supply and demand constraints when planning new EV charging stations	Beijing	EV ranges Population Govt budget constraints	N	N	Set and Max Covering P-median	Best solution satisfying restrictions
[9]	Using allocation objectives to plan EV infrastructure	Hong Kong	EV usage rate survey	N	N	Location-Allocation	Compared to standard model
[10]	Spatial and temporal information to predict EV station availability	Dundee, Ireland	EV infrastructure	N	Spatial Temporal CGN	N	RMSE MAE R^2
[11]	Predictive model for estimating the EV infrastructure usage	Netherlands	ELaadNL Google Maps Nokia HERE	LinReg XGBoost	N	N	RMSE MAE
[12]	Evaluate factors affecting EV infrastructure popularity	Netherlands	EVnetNL OpenGIS EV Charging Records	LogReg Rand Forest GBRT	N	N	Accuracy Precision Sensitivity MCC
[13]	Predict energy demand for next 24 hours at each charging outlet	UCLA	EV Charging Records	KNN ARIMA	N	Pattern Sequence Forecasting	Accuracy
[14]	EV Infrastructure planning based on user travel data	Shanghai	Shanghai International Motor City	N	N	Mixed-Integer Nonlinear	Best model given constraints
[15]	Bridge the gap for EV access between the wealthy and the poor	Orange County	CEC Energy Dept U.S. Census	Rand Forest Multi-Log Regression	N	N	Precision Recall F1-score
[16]	Predict charging demand and EV infrastructure placement	Georgia Tech	Charging station usage	N	EMD- AOA- LSTM	N	MAE RMSE Accuracy
[17]	Data-driven long-term EV infrastructure planning	Sydney	Traffic Flow Distribution Network	N	CNN	Cournot Competition Model	
[18]	Models classifying EV charging user reviews	United States	Charging station user reviews	N	N	BERT XLNet	Accuracy F1 score
[19]	Predicting charging duration at public chargers	Rabat, Morocco	Real-time EV station	N	RNN LSTM GRU	N	MSE RMSE MAE
[20]	Predicting PEV charging demand	Nebraska	Level 2 EV stations	LinReg GBRT SVM	N	N	RMSE MAE R^2
[21]	Infrastructure planning for electric taxi vehicles	Vienna	Taxi dispatch center	N	N	Decision Support System	
[22]	Describe the urban context and activities near EV infrastructure	Netherlands	EVnetNL	LogReg Rand Forest GBRT	N	N	Accuracy Precision Sensitivity MCC
[23]	Minimize the total charging cost of EVs	Singapore	LTA Stats Dept	N	N	CSPP Nash Transfer	Most EVs Fit Budget
[24]	Comparing 3 ML models to understand EV charging demand	Agartala, India	Traffic Flow	CG - SMV Bagged and Boosted Tree	N	N	RMSE MAE R^2
[25]	Predict charging station availability using ML Big Data	Italy	EV Charge Transactions	Logistic Regression	N	N	Precision Recall F1 score
[26]	Predicting charging times for different EV charging modes	Japan	Auto Research Institute	Ensemble Rand Forest Grad Boost	N	N	RMSE MAE R^2
[27]	EV planning from the perspective of private investor	China	Transport & Sioux-Fall Networks	Mixed-Integer NLP Huff model	N	N	Highest investor profit

TABLE III
APPROACHES TO PLAN FUTURE EV INFRASTRUCTURE

Ref	Objective	Methods	Description	Negatives
[30]	Methods used for locating charging infrastructure	Node-based approach	Locates the charging points/ stations to meet the area's overall demand at the given instance.	Does not always leads to the best coverage with the minimal cost
		Path-based approach	Based on vehicle paths given origin and destination of trips. Have to make a stop in between trips for charging.	Approach is worse when predicting slow charging stations
		Tour-based approach	More likely to reflect the real charging needs of EV users. Take into account the whole journey of EV.	Requires detailed individual travel data and high computing capacity
[31]	Planning EV Infrastructure for different types of vehicles	Buses only	Easy to track bus location and less chance of excess charge being consumed because of their fixed routes	Routes are long and buses must charge during down time
		Taxis only	Converting taxis to be fully EVs makes the most sense because they use a lot of fuel to mostly travel short distances	Must charge between trips Losing money if charging time is too long
		All Private Vehicles	Most realistic method because it includes data from all vehicles that take part in traffic	Requires lot of data - battery capacities, charging behaviors, etc. of each EV
	Maximize Charging Infrastructure Utilization	Infinite size shrinks to demand-based	Initially assume that location can accommodate infinite stations. Once location has been found, shrink station size according to demand	Size of station influences station location - incorrect assumption at start will lead to improper planning
		Station size constraint provided at start	Some models add a constraint of station size to better serve areas of high charging demand concentration with multiple stations.	Charging station size constraint at the start doesn't allow for flexibility of station location
		Power sizing	Important to choose station power levels before modeling. Must take into account grid capacity at location of charging points to avoid overloading	None provided

in the location. There are some parameters related to the amount and types of EVs to support in the location that can be modified on the results page before recalculating the results.

We also did some research on a couple of methods that are used to plan EV infrastructure in a region which are listed in Table III. The first paper talks about methods used to plan for locating where future EV infrastructure would go. It describes the three different approaches used for locating charging infrastructure. The second paper talks about the EV infrastructure planning with respect to the type of vehicles and how to utilize the charging to its maximum level.

III. DATA ENGINEERING

This section gives a comprehensive explanation of the data collection and data pre-processing which includes Data Cleaning and Data Merging.

A. Data Collection

The primary focus of this research is on the state of California, specifically in the cities and zip codes of San Jose and Palo Alto. We have decided to use three different datasets to solve our problems. The first dataset is from the Alternative Fuels Data Center website from the US Department of Energy.

This dataset contains information about charging stations and their locations throughout the state of California, such as latitude, longitude, address, and charger type. The second dataset is the vehicle population dataset by zip code that we obtained from the California Energy commission. We used this dataset to enhance the first dataset in order to understand how the number of electric vehicles on the road affects the number of EV charging stations available in the zip code. The third dataset that we obtained is from the Palo Alto government and it contains detailed data for a few EV charging stations in Palo Alto, including charging transactions, the plug-in time of the Electric vehicles, and the amount of Energy that is consumed by each charging transaction.

More information about these datasets can be found in the table below.

The dataset is of real-time data and is updated with the latest information. The data has been extracted through api format and the dataset has been downloaded in CSV format. This dataset has been geocoded by the latitude and longitude details of the charging station details.

B. Dataset Preprocessing

Data preprocessing is an important step which involves cleaning and organizing the raw data collected so that it is

TABLE IV
DATASET COLLECTION

	San Jose	Vehicle	Palo Alto
Dataset	EV Charging stations data	population - zip codes	EV Charging Station Usage data
Data type	csv	xlsx	csv
Size	15.7 MB	4.4 MB	13.3MB
No of rows	14444	172582	259,443
Yearly	Y	Y	Y
Weekly	N	N	Y
Monthly	N	N	Y
Duration	2010 -2022	2010-2021	Real-Time

more suitable for feeding into the model[34]. In our project the dataset is real-time containing null, irrelevant and missing data.

The following steps are performed to refine the raw dataset

a) *Data Cleaning*: This step takes care of dropping null and unnecessary data, handling missing values, renaming the columns, filtering the data with respect to California and finally processing the datasets.

b) *Data Merging*: In this stage, all the datasets are merged to the EV station dataset based on various parameters. The vehicle registration data is merged based on the zip code. EV population dataset is merged based on the zip code and opening year. After merging all the datasets, the final dataset is processed, and it consists of 32 columns and 10k rows.

The below pie chart here represents the percentage of number of features in every dataset which can be used for model building.

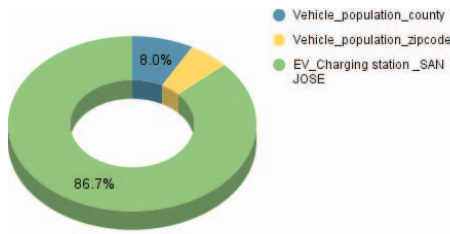


Fig. 1. Number of features in refined datasets

IV. MACHINE LEARNING MODELS

This section gives an in detail description of different types of machine learning models used for the given problem statement.

A. SARIMAX

The SARIMAX model is an extension of the ARIMA model that includes seasonality. When patterns are not consistent but occur periodically, there is a seasonality in the series. There are parallel time series that are not modeled directly but can be used as a weighted input known as an exogenous variable. The

disadvantage of the ARIMA model is that it cannot include seasonality and exogenous variables. By including an additional set of autoregressive and moving average components a SARIMA model is derived. There are additional lags from the offset by the frequency of seasonality. The SARIMAX model is an updated version of the ARIMA model [35].

The SARIMA model has the following four orders: a seasonal autoregressive order (P), an order of seasonal integration (D), a seasonal moving average order (Q), and a seasonal order, which is the length of the cycle of repeating frequency. The Seasonal order is given by four numbers: the seasonal AR specification, the seasonal Integration order, the seasonal MA, and the seasonal periodicity. SARIMAX is a variation of SARIMA where it considers the exogenous variable, which is the external data of forecast. The following figure shows the equation of the SARIMAX model.

$$d_t = c + \sum_{n=1}^p \alpha_n d_{t-n} + \sum_{n=1}^q \theta_n \epsilon_{t-n} + \sum_{n=1}^r \beta_n x_n + \sum_{n=1}^P \phi_n d_{t-sn} + \sum_{n=1}^Q \eta_n \epsilon_{t-n} + \epsilon_{t-n}$$

Fig. 2. Sarimax Model Equation

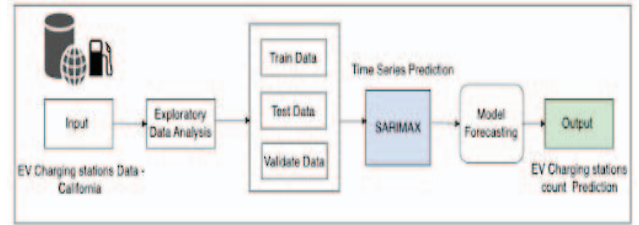


Fig. 3. Sarimax Model

B. Meta Fusion Regression Model

It is an integrated model [36] which is based on the stacked model and consists of the traditional machine learning models like Random Forest Regression, XGBoost and Decision Tree Regression models. The input data is given to the regressor model, which predicts the energy consumption as the output.

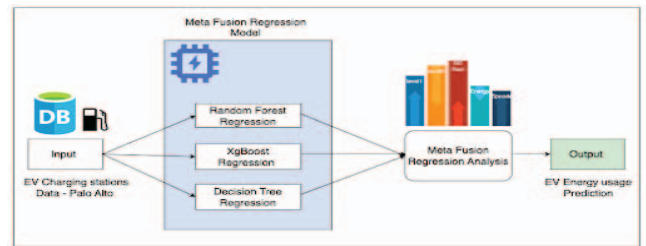


Fig. 4. Meta Fusion Regressor Model

C. K - Means Clustering Recommendation model

The research paper presents a data-driven approach for the planning of electric vehicle (EV) charging infrastructure in a smart city. The approach uses K-Means clustering [37] and geographical visualization tools to identify clusters of high EV traffic and low charging station density and calculates the number of charging stations needed in each cluster based on historical charging data. The results are used to optimize the deployment of charging stations in the identified clusters.

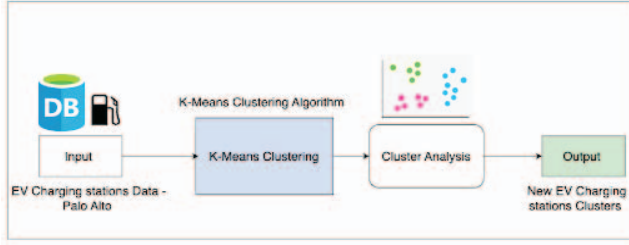


Fig. 5. K Means Clustering

V. RESULTS AND DISCUSSION

This section primarily focuses on the results obtained after applying machine learning models. We also discuss and compare the results being obtained.

A. EV charging station count prediction

Since the load dataset contains monthly data and they contain information based on zip code, we must separate the data by zip code and then run the models on each zip code. We decided to run the models on the top 10 most populous zip codes to understand how the models work. For all the models, we split each zip code dataset into 80% training data and 20% test data. The following figure shows a visualization of how we split the original dataset into test and training sets. We used the SARIMAX model to predict the load for each zip code. This model is one of the most complex models yet is the one with the best prediction accuracy.

For this model, we used the Auto Arima function from the “pmdarima” library to predict the values of the parameters that we would use in the model. The result of the Auto Arima function is shown in the following figure. We plugged in the parameters into our SARIMAX model to run it and predict the test data.

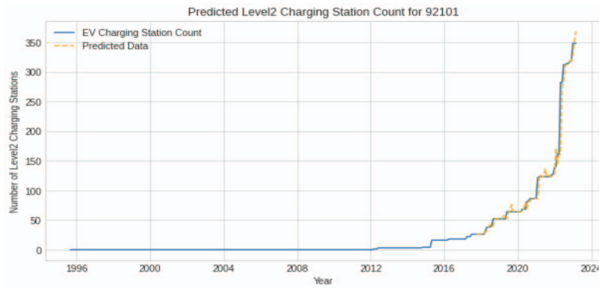


Fig. 6. Test and Prediction value results for 92101 Zipcode

The above graph shows the test and prediction values using this model and Table V shows the results of evaluating the model.

TABLE V
EVALUATION OF EV CHARGING STATION COUNT PREDICTION

Model	RMSE	MAE	R ²
SARIMAX	17.43	6.50	0.967

B. Predicting EV Energy Usage

This task mainly focuses on how much energy an EV charging station consumes. This is important because we need to advise our users to understand the consequences of building a new charging station. One of the major considerations when building new EV stations is to figure out how much energy is being consumed from the power grid. If an area of town is already consuming too much electricity, we would recommend that they should build their station at a different location. We used the Palo Alto EV station dataset for this problem. The dataset also has a feature, which is the amount of charging time that each transaction takes. We must run our models on individual charging stations to predict its energy consumption because we do not want data from different stations to interfere with our models. For this task we introduce our new model called Meta Fusion Regressor which combines the Random Forest, XGBoost, Decision Tree and other traditional Machine Learning models. We ran the model on different time periods - monthly, weekly and daily - to see the differences between the model accuracies and the energy consumption in each period. For this prediction, we also split the dataset by using the earliest 80% of the data for one station as the training dataset and the remaining 20% as the test dataset. The following figure shows how we split the dataset after we prepared it for weekly

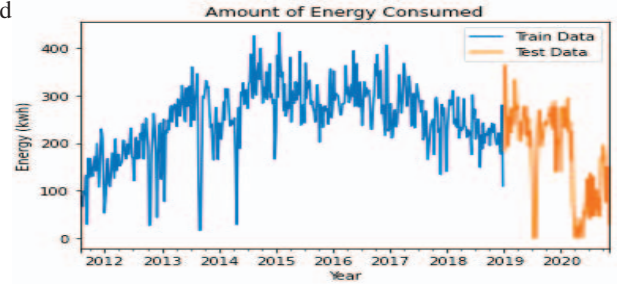


Fig. 7. Splitting of train and test dataset

The dataset has a feature, which is the amount of charging time that each transaction takes. This feature is helpful in predicting the amount of energy used because as the time taken to charge the electric vehicle increases, the more energy is consumed from the grid. Fig 8, 9 and 10 represent the test and prediction results of the model.

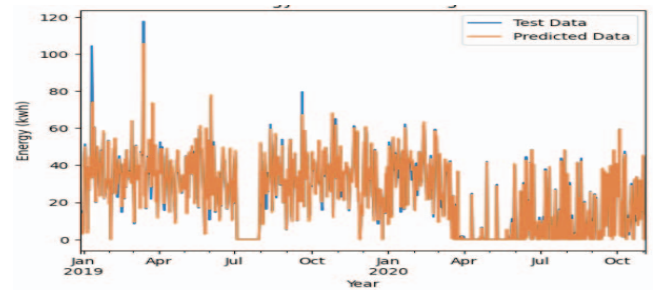


Fig. 8. Test and Prediction data results on Daily Basis

Table VI shows that the Meta Fusion Regressor has the lowest RMSE and MAE values and the highest R2 score across

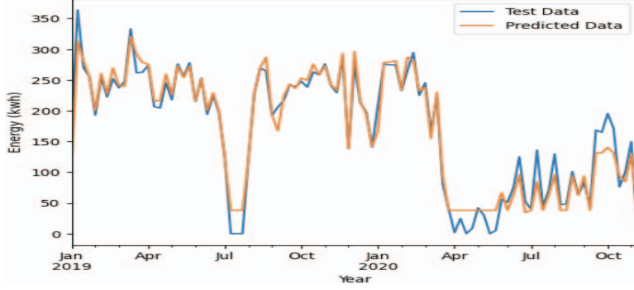


Fig. 9. Test and Prediction data results on Weekly Basis

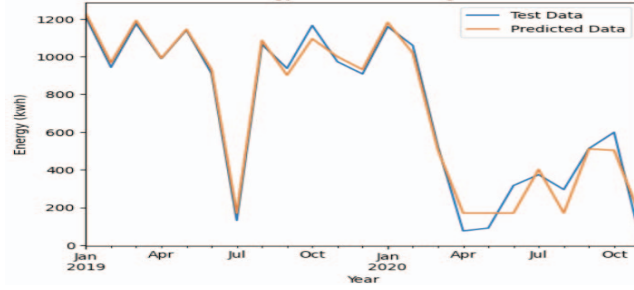


Fig. 10. Test and Prediction data results on Monthly Basis

all of the time periods tested. This means that the values that the meta fusion regression predicted were the closest to the test values compared to the other models. A higher R^2 score means that the predictions that the meta fusion regressor makes fits the dataset better than the other models.

TABLE VI
MODEL COMPARISON FOR PREDICTING THE EV ENERGY USAGE

Timeframe	Model	R^2 Score	MAE	RMSE
Daily	Meta Fusion Regressor	0.98	0.91	2.26
	Random Forest	0.98	0.99	2.47
	XG Boost	0.98	0.81	2.65
	Decision Tree	0.97	0.90	3.34
	Meta Fusion Regressor	0.95	14.30	20.59
Weekly	Random Forest	0.89	18.93	30.58
	XG Boost	0.93	16.06	24.14
	Decision Tree	0.87	20.58	34.29
	Meta Fusion Regressor	0.97	46.39	62.10
	Random Forest	0.68	145.54	225.60
Monthly	XG Boost	0.87	88.10	140.86
	Decision Tree	0.14	245.24	373.08

C. Optimal EV Charging station Location Prediction

The final problem that we performed is to give a recommendation of the new optimal EV charging station's locations. This is important because as there is an increase in the adoption for electric vehicles, there should be an increase in the number of EV charging stations to meet the EV

charging demand. We have taken the historical EV charging station data of Palo Alto and applied the K-means clustering algorithm to identify the clusters of high EV traffic and low EV charging station density.

The centroids of each individual cluster are identified, and the count of EV charging stations needed is calculated by dividing the total EV charging time by the number of unique EV charging stations identified in that cluster. The obtained number is then rounded up to the nearest integer. The results obtained are plotted over a Palo-Alto centered map which was created using the Folium library.

As shown in the below figure, the current EV charging stations are displayed using green icons, and the predicted EV charging stations are displayed using red icons.

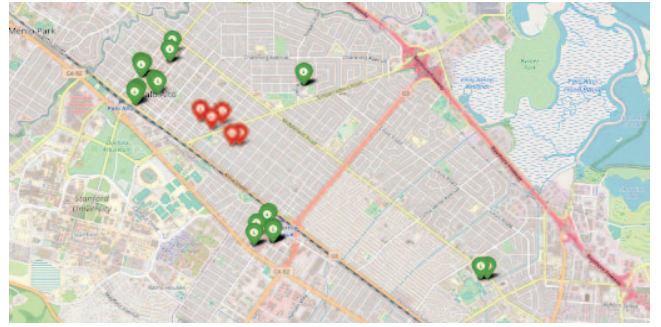


Fig. 11. Optimal EV Charging station Location Prediction

VI. CONCLUSION

This paper presents three different approaches to address the challenges of planning new EV charging station infrastructure. The first challenge that was tackled is predicting the number of EV charging stations that should be placed in a particular zip code in the next three years. This was done by calculating the historical number of EV charging stations in the zip code and then using the SARIMAX model to forecast the future counts. The second challenge which was addressed is predicting how much energy (kWh) is consumed by a charging station in daily, weekly, and monthly time periods based on different weather conditions. The proposed stacked model, called Meta Fusion Regression, provided the best prediction results. The third challenge utilized the K-means clustering algorithm to predict the optimal locations for EV charging stations.

Overall, the above approaches helped to provide valuable information in planning new EV charging infrastructure in a zip code. In the future, live traffic data can be incorporated into the approaches in order to provide better understanding of how drivers behave when they want to charge their electric vehicles.

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